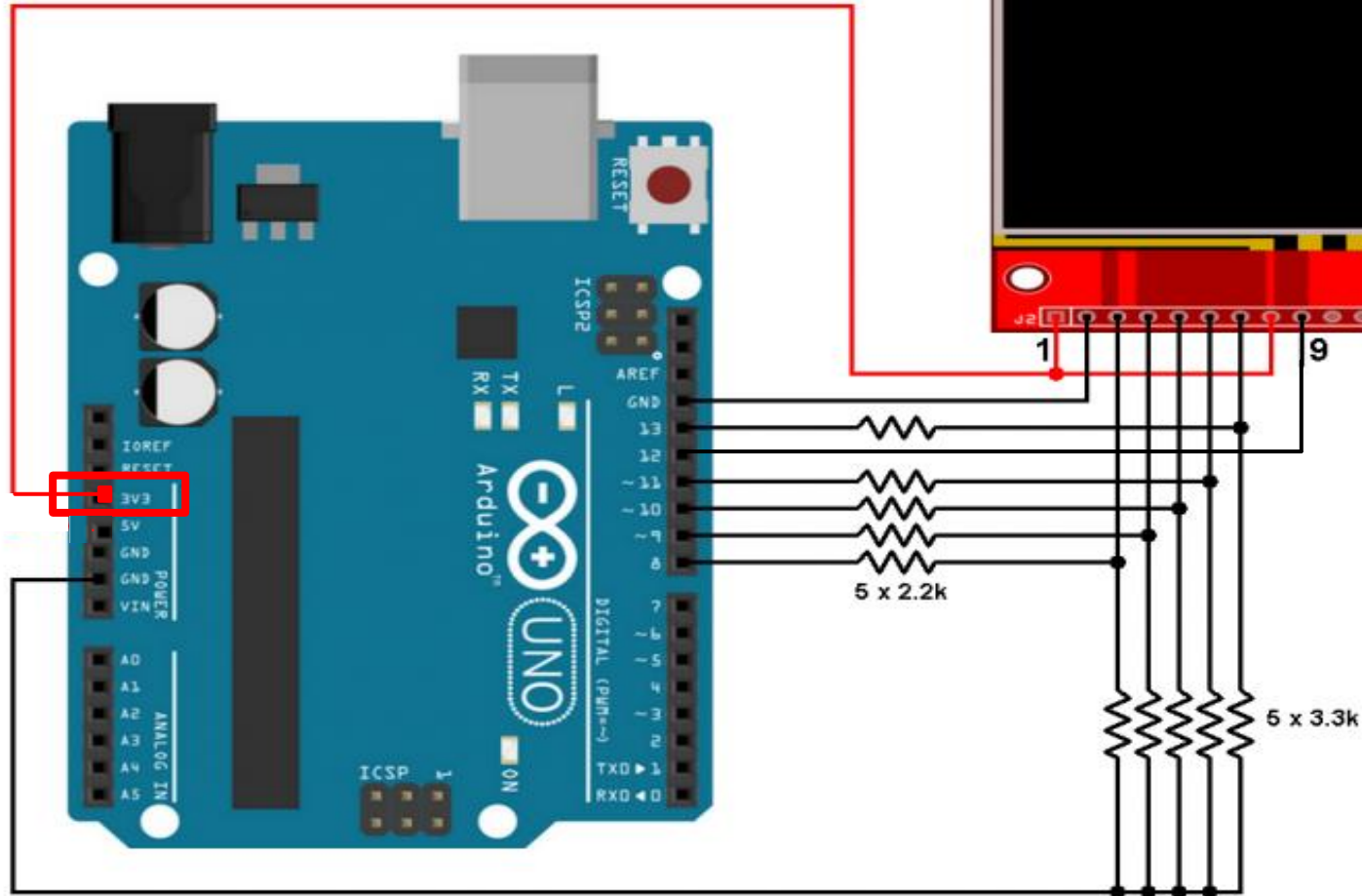
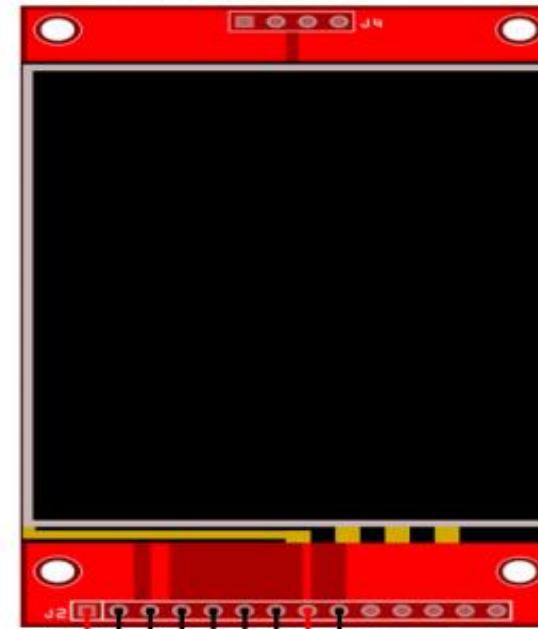


TFT Pinout:

1	2	3	4	5	6	7	8	9
VCC	GND	CS	RST	D/C	MOSI	SCK	BL	MISO

ILI9341 TFT



注意：
1.VCC、BL(LED)請
接3.3V，以免損壞
LCD板

Serial.print(F("Hello World"));

- ▶ When you compile your program it says
 - ▶ (1) how much program memory (stored in flash) you are using and
 - ▶ (2) how much dynamic ram you are using.
- ▶ If you use F() you can move constant strings to the **program memory** instead of the ram.
- ▶ the **compiled string stays in flash** and does not get copied to SRAM during the C++ initialization that happens before your sketch receives run control.
- ▶ Since the string is not moved to SRAM, it has the **PROGMEM** property and runs from flash.

- ▶ Your string constants are ALWAYS in flash memory.
- ▶ They have to be, because when you take power away from RAM everything's gone.
- ▶ When you put power back on again, they have to be reloaded from somewhere, and that somewhere is the flash memory.
- ▶ Leaving off PROGMEM will do nothing to save you significant amounts of flash memory.
- ▶ It's just that a normal part of initializing a C++ program is to load those arrays into SRAM before setup().
- ▶ PROGMEM is an AVR-specific attribute that can be applied to variables that tells the compiler to not do that.
- ▶ Because flash is in a different memory space than RAM, special functions are needed to access from flash than from RAM.

腳位說明

符號	腳位名稱	說明
VCC	邏輯電源	接電源正
GND	邏輯地	接電源地
CS	片選訊號	標準的SPI Pin
RST	RESET	重設LCD
D/C	Data/Command	在Low的時候, 表示此時SPI傳送的都是命令, 而在High的時候表示傳送的都是資料
MOSI	主機輸出從機輸入訊號 (資料由主機發出)	標準的SPI Pin
SCK	串列時脈	標準的SPI Pin
BL(LED)	背光電源	需要有一定的亮度才可以看到螢幕內容, 通常可用PWM控制, 或直接接到VCC讓它100%發亮
MISO	主機輸入從機輸出訊號 (資料由從機發出)	標準的SPI Pin

範例程式

初始設定 - 匯入函式庫、定義pin腳位

- ▶ `#include "SPI.h"`
- ▶ `#include "Adafruit_GFX.h"`
- ▶ `#include "Adafruit_ILI9341.h"`

- ▶ `#define TFT_CLK 13`
- ▶ `#define TFT_MISO 12`
- ▶ `#define TFT_MOSI 11`
- ▶ `#define TFT_DC 9`
- ▶ `#define TFT_CS 10`
- ▶ `#define TFT_RST 8`
- ▶ `//建立LCD class`
- ▶ `Adafruit_ILI9341 tft = Adafruit_ILI9341(TFT_CS, TFT_DC, TFT_MOSI, TFT_CLK, TFT_RST, TFT_MISO);`

範例程式

初始設定 - 設定序列、LCD

```
▶ void setup() {  
▶   Serial.begin(9600);  
▶   Serial.println("ILI9341 Test!");  
▶   tft.begin();  
▶   // 若輸出皆為0x0，請RESET arduino並確認接線是否有誤  
▶   // read diagnostics (optional but can help debug problems)  
▶   uint8_t x = tft.readcommand8(ILI9341_RDMODE);  
▶   Serial.print("Display Power Mode: 0x"); Serial.println(x, HEX);  
▶   x = tft.readcommand8(ILI9341_RDMADCTL);  
▶   Serial.print("MADCTL Mode: 0x"); Serial.println(x, HEX);  
▶   x = tft.readcommand8(ILI9341_RDPIXFMT);  
▶   Serial.print("Pixel Format: 0x"); Serial.println(x, HEX);  
▶   x = tft.readcommand8(ILI9341_RDIMGFMT);  
▶   Serial.print("Image Format: 0x"); Serial.println(x, HEX);  
▶   x = tft.readcommand8(ILI9341_RDSELDIAG);  
▶   Serial.print("Self Diagnostic: 0x"); Serial.println(x, HEX);  
▶ }
```

範例程式

範例開始 - 依序調用function

```
Serial.println(F("Benchmark      Time (microseconds)"));
delay(10);
Serial.print(F("Screen fill      "));
Serial.println(testFillScreen());
delay(500);

Serial.print(F("Text              "));
Serial.println(testText());
delay(3000);

Serial.print(F("Lines             "));
Serial.println(testLines(ILI9341_CYAN));
delay(500);

Serial.print(F("Horiz/Vert Lines          "));
Serial.println(testFastLines(ILI9341_RED, ILI9341_BLUE));
delay(500);

Serial.print(F("Rectangles (outline)        "));
Serial.println(testRects(ILI9341_GREEN));
```

```
delay(500);

Serial.print(F("Rectangles (filled)      "));
Serial.println(testFilledRects(ILI9341_YELLOW, ILI9341_MAGENTA));
delay(500);

Serial.print(F("Circles (filled)              "));
Serial.println(testFilledCircles(10, ILI9341_MAGENTA));

Serial.print(F("Circles (outline)            "));
Serial.println(testCircles(10, ILI9341_WHITE));
delay(500);

Serial.print(F("Triangles (outline)          "));
Serial.println(testTriangles());
delay(500);

Serial.print(F("Triangles (filled)           "));
Serial.println(testFilledTriangles());
delay(500);
```

```
Serial.print(F("Rounded rects (outline)    "));
Serial.println(testRoundRects());
delay(500);

Serial.print(F("Rounded rects (filled)      "));
Serial.println(testFilledRoundRects());
delay(500);

Serial.println(F("Done!"));
}
```

- ▶ 加上 F(), 當中的 string 會放到 flash memory 中, 而不會佔用 SRAM。

範例程式

testFillScreen()

```
► unsigned long testFillScreen() {  
►   unsigned long start = micros();  
►   tft.fillScreen(ILI9341_BLACK);  
►   yield();  
►   tft.fillScreen(ILI9341_RED);  
►   yield();  
►   tft.fillScreen(ILI9341_GREEN);  
►   yield();  
►   tft.fillScreen(ILI9341_BLUE);  
►   yield();  
►   tft.fillScreen(ILI9341_BLACK);  
►   yield();  
►   return micros() - start;  
► }
```

- **micros()** : 此函式會傳回從開機到現在的微秒數, 傳回值為 **unsigned long**。
- **tft.fillScreen(uint16_t color)** : 以 **color** 填滿畫面。
- 顏色的計算請參考(程式碼說明部分)<http://mt7688jackychi.blogspot.com/2016/08/ameba-arduino-spi-lcd.html>

範例程式

testText()

```
▶ unsigned long testText() {  
▶   tft.fillScreen(ILI9341_BLACK);  
▶   unsigned long start = micros();  
▶   tft.setCursor(0, 0);           //設定游標位置  
▶   tft.setTextColor(ILI9341_WHITE); //設定文字顏色  
▶   tft.setTextSize(1);           //設定文字大小  
▶   tft.println("Hello World!");   //顯示文字  
▶   tft.setTextColor(ILI9341_YELLOW); tft.setTextSize(2);  
▶   tft.println(1234.56);  
▶   tft.setTextColor(ILI9341_RED);  tft.setTextSize(3);  
▶   tft.println(0xDEADBEEF, HEX);    //也能輸出數字  
▶   tft.println();  
▶   tft.setTextColor(ILI9341_GREEN);  
▶   return micros() - start;  
▶ }
```

範例程式

testLines()

```
▶ unsigned long testLines(uint16_t color) {  
▶   unsigned long start, t;  
▶   int          x1, y1, x2, y2, w = tft.width(), h = tft.height();  
▶   tft.fillScreen(ILI9341_BLACK); // 清除畫面  
▶   yield();  
▶   x1 = y1 = 0;  
▶   y2  = h - 1;  
▶   start = micros();  
▶   for(x2=0; x2<w; x2+=6) tft.drawLine(x1, y1, x2, y2,  
▶   color);  
▶   x2  = w - 1;  
▶   for(y2=0; y2<h; y2+=6) tft.drawLine(x1, y1, x2, y2,  
▶   color);  
▶   t    = micros() - start; // fillScreen doesn't count against  
▶   timing  
▶   yield();  
▶   return micros() - start;  
▶ }
```

- ▶ tft.drawLine(int16_t x1, int16_t y1, int16_t x2, int16_t y2, uint16_t color)
 - ▶ 從(x1,y1)畫線至(x2,y2)
 - ▶ (0,0)在左上角

範例程式

testFastLines()

- ▶ `unsigned long testFastLines(uint16_t color1, uint16_t color2) {`
- ▶ `unsigned long start;`
- ▶ `int x, y, w = tft.width(), h = tft.height();`
- ▶ `tft.fillScreen(ILI9341_BLACK); // 清除畫面`
- ▶ `start = micros();`
- ▶ `for(y=0; y<h; y+=5) tft.drawFastHLine(0, y, w, color1);`
- ▶ `for(x=0; x<w; x+=5) tft.drawFastVLine(x, 0, h, color2);`
- ▶ `return micros() - start;`
- ▶ `}`
- ▶ `tft.drawFastHLine(int16_t x, int16_t y, int16_t w, uint16_t color)`
 - ▶ 由(x,y)畫長w的水平線
- ▶ `tft.drawFastVLine(int16_t x, int16_t y, int16_t w, uint16_t color)`
 - ▶ 由(x,y)畫長w的垂直線

範例程式

testRects()

- ▶ unsigned long testRects(uint16_t color) {
- ▶ unsigned long start;
- ▶ int n, i, i2, cx = tft.width() / 2,
- ▶ cy = tft.height() / 2;
- ▶ tft.fillScreen(ILI9341_BLACK);
- ▶ n = min(tft.width(), tft.height());
- ▶ start = micros();
- ▶ for(i=2; i<n; i+=6) {
- ▶ i2 = i / 2;
- ▶ tft.drawRect(cx-i2, cy-i2, i, i, color);
- ▶ }
- ▶ return micros() - start;
- ▶ }
- ▶ tft.drawRect(int16_t x, int16_t y, int16_t w, int16_t h, uint16_t color)
- ▶ 以(x,y)為左上角，畫出w*h的空心矩形。

範例程式

testFilledRects()

```
▶ unsigned long testFilledRects(uint16_t color1, uint16_t color2)
{
▶   unsigned long start, t = 0;
▶   int          n, i, i2, cx = tft.width() / 2 - 1, cy = tft.height() /
2 - 1;
▶   tft.fillScreen(ILI9341_BLACK);
▶   n = min(tft.width(), tft.height());
▶   for(i=n; i>0; i-=6) {
▶     i2  = i / 2;
▶     start = micros();
▶     tft.fillRect(cx-i2, cy-i2, i, i, color1);
▶     t  += micros() - start; // Outlines are not included in
timing results
▶   tft.drawRect(cx-i2, cy-i2, i, i, color2);
▶   yield();
▶   }
▶   return t;
▶ }
```

▶ tft.fillRect(int16_t x, int16_t y, int16_t w, int16_t h, uint16_t color)

▶ 以(x,y)為左上角，畫出w*h的實心矩形。

範例程式

testFilledCircles()

- ▶ unsigned long testFilledCircles(uint8_t radius, uint16_t color) {
- ▶ unsigned long start;
- ▶ int x, y, w = tft.width(), h = tft.height(), r2 = radius * 2;
-
- ▶ tft.fillScreen(ILI9341_BLACK); // 清除畫面內容
- ▶ start = micros();
- ▶ for(x=radius; x<w; x+=r2) {
- ▶ for(y=radius; y<h; y+=r2) {
- ▶ tft.fillCircle(x, y, radius, color);
- ▶ }
- ▶ }
- ▶ return micros() - start;
- ▶ }

- ▶ tft.fillCircle(int16_t x0, int16_t y0, int16_t r, uint16_t color)
- ▶ (x0,y0)為圓心，半徑為r的實心圓

範例程式

testCircles()

- ▶ unsigned long testCircles(uint8_t radius, uint16_t color) {
- ▶ unsigned long start;
- ▶ int x, y, r2 = radius * 2, w = tft.width() + radius,
- ▶ h = tft.height() + radius;
- ▶ // Screen is not cleared for this one -- this is
- ▶ // intentional and does not affect the reported time.
- ▶ start = micros();
- ▶ for(x=0; x<w; x+=r2) {
- ▶ for(y=0; y<h; y+=r2) {
- ▶ tft.drawCircle(x, y, radius, color);
- ▶ }
- ▶ }
- ▶ return micros() - start;
- ▶ }
- ▶ tft.drawCircle(int16_t x0, int16_t y0,
- ▶ int16_t r, uint16_t color)
- ▶ (x0,y0)為圓心，半徑為r的空心圓

範例程式

testTriangles()

```
▶ unsigned long testTriangles() {  
▶   unsigned long start;  
▶   int          n, i, cx = tft.width() / 2 - 1,  
▶               cy = tft.height() / 2 - 1;  
▶   tft.fillScreen(ILI9341_BLACK);  
▶   n = min(cx, cy);  
▶   start = micros();  
▶   for(i=0; i<n; i+=5) {  
▶     tft.drawTriangle(  
▶       cx, cy - i, // peak  
▶       cx - i, cy + i, // bottom left  
▶       cx + i, cy + i, // bottom right  
▶       tft.color565(i, i, i));  
▶   }  
▶   return micros() - start;  
▶ }
```

- ▶ tft.drawTriangle(int16_t x0, int16_t y0, int16_t x1, int16_t y1, int16_t x2, int16_t y2, uint16_t color)
 - ▶ (x0,y0)上方點
 - ▶ (x1,y1)左下點
 - ▶ (x2,y2)右下點
 - ▶ 此為空心三角形

範例程式

testFilledTriangles()

```
▶ unsigned long testFilledTriangles() {  
▶   unsigned long start, t = 0;  
▶   int          i, cx = tft.width() / 2 - 1, cy = tft.height() / 2  
- 1;  
▶   tft.fillScreen(ILI9341_BLACK);  
▶   start = micros();  
▶   for(i=min(cx,cy); i>10; i-=5) {  
▶     start = micros();  
▶     tft.fillTriangle(cx, cy - i, cx - i, cy + i, cx + i, cy + i,  
▶       tft.color565(0, i*10, i*10));  
▶     t += micros() - start;  
▶     tft.drawTriangle(cx, cy - i, cx - i, cy + i, cx + i, cy + i,  
▶       tft.color565(i*10, i*10, 0));  
▶     yield();  
▶   } return t;  
▶ }
```

- ▶ tft.fillTriangle(int16_t x0, int16_t y0, int16_t x1, int16_t y1, int16_t x2, int16_t y2, uint16_t color)
 - ▶ (x0,y0)上方點
 - ▶ (x1,y1)左下點
 - ▶ (x2,y2)右下點
 - ▶ 此為實心三角形

範例程式

testFilledRoundRects()

- ▶ unsigned long testFilledRoundRects() {
- ▶ unsigned long start;
- ▶ int i, i2, cx = tft.width() / 2 - 1, cy = tft.height() / 2 - 1;
- ▶ tft.fillScreen(ILI9341_BLACK);
- ▶ start = micros();
- ▶ for(i=min(tft.width(), tft.height()); i>20; i-=6) {
- ▶ i2 = i / 2;
- ▶ tft.fillRoundRect(cx-i2, cy-i2, i, i, i/8, tft.color565(0, i, 0));
- ▶ yield();
- ▶ }
- ▶ return micros() - start;
- ▶ }

- ▶ tft.fillRoundRect(int16_t x0, int16_t y0, int16_t w, int16_t h, int16_t radius, uint16_t color)
 - ▶ 以(x0,y0)為左上角，畫出w*h的實心圓角矩形。

影片

- ▶ <https://youtu.be/f-T9fHrFrIk?t=35>

